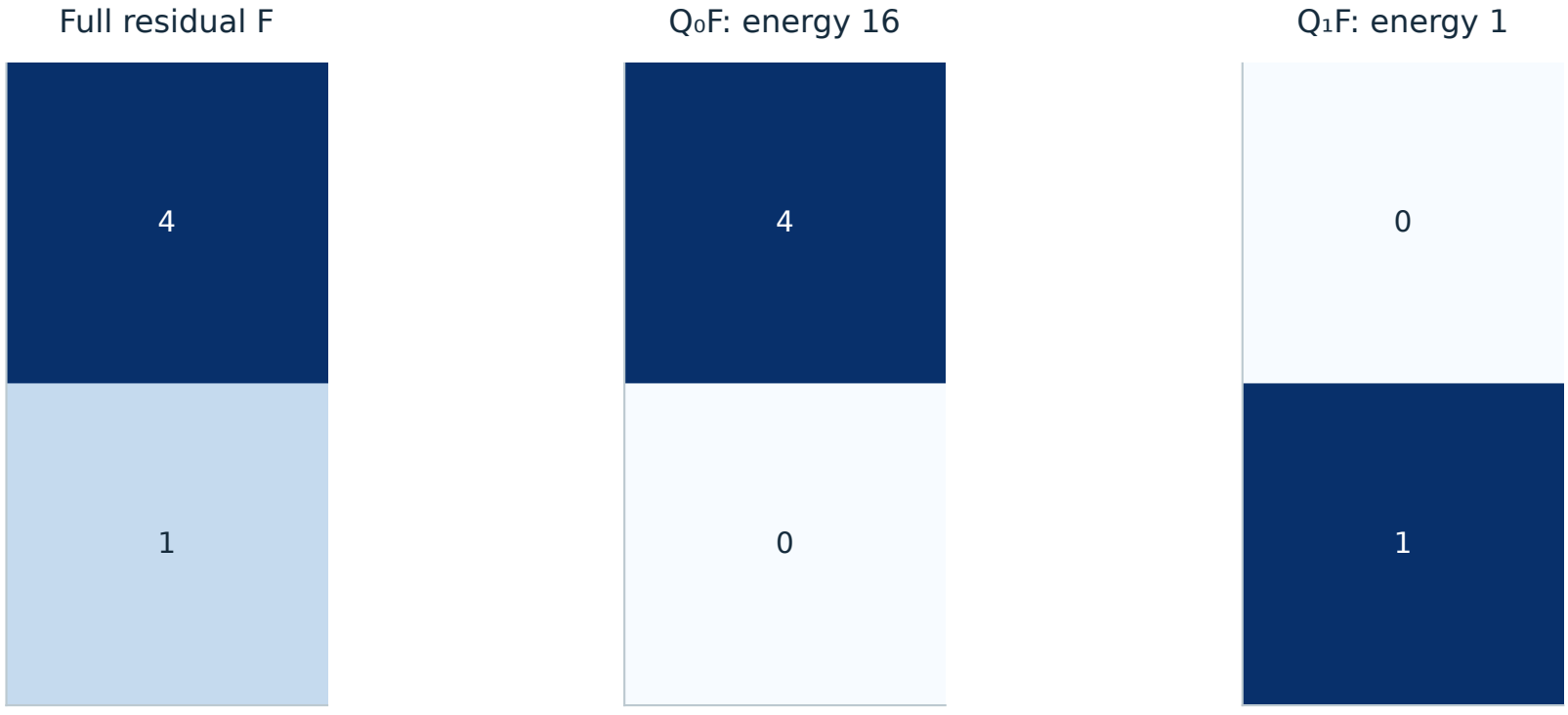
Packets account for the whole residual

At $B_0 = 0$, $F = PH^T = (4, 1)^T$. Orthogonal packets split the energy.



$$\|F\|^2 = 16 + 1 = 17$$

Exact matrices and exact energy identity at the coordinate initial gain.